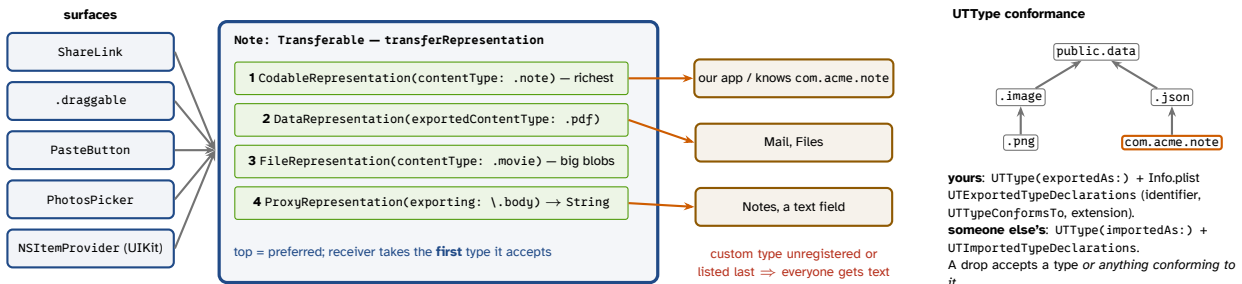


In one line: **Transferable** (iOS 16) describes **once**, as an **ordered list of representations keyed by UTType**, how a value leaves and enters the app; **ShareLink**, **.draggable**, **.dropDestination**, **PasteButton** and **PhotosPicker** all consume it — replacing separate **UIActivityItemSource**, **NSItemProvider** and **UIPasteboard** code.



How it works

- **Representations** (result builder, order = preference): **Codable** (JSON by default); **Data** (exporting/importing: closures, whole blob in RAM); **File** (a file URL, lazy — export returns `SentTransferredFile`, import gets `ReceivedTransferredFile` whose `.file` is **temporary** — **copy it before returning**); **Proxy** (reuse String, URL, Data, Image). Modifiers: `.suggestedFileName`, `.exportingCondition`.
- **Drag & drop**: `.draggable(item) { preview }`; `.dropDestination(for: T.self) { items, location in ... return true }` is `isTargeted: { }`. Only offered types that match T arrive.
- **Share**: `ShareLink(item:subject:message:preview: SharePreview(title, image:))`.
- **Paste**: `PasteButton(payloadType:)` — the tap is consent, so no “Allow Paste” alert (iOS 16 alerts on programmatic reads of another app's content; UIKit: `UIPasteControl`).
- **PhotosPicker(selection:matching:)** (16) runs **out of process** — no photo-library permission; the app sees only picked items. `await item.loadTransferable(type:)` (in `.task(id: item)`) returns `T?`: `nil` = type not offered, throws = load failed. Video → your own `FileRepresentation` type.
- **Files**: `.fileImporter` hands back **security-scoped** URLs outside your sandbox; `.fileExporter` writes a `FileDocument` (17 adds a `Transferable` form).
- **UIKit bridge**: `UIDragItem(itemProvider:)`, `UIDropInteraction`, `UIActivityViewController`. `NSItemProvider` gained `register(_:)` for a `Transferable` and `loadTransferable(type:)` (16); older SwiftUI: `.onDrag` / `.onDrop(of:)`.

Example — one type, every surface

```
extension UTType {
    static let note = UTType(exportedAs: "com.acme.note")
}
struct Note: Codable, Transferable {
    var title: String; var body: String
    static var transferRepresentation: some TransferRepresentation {
        CodableRepresentation(contentType: .note) // richest first
        ProxyRepresentation(exporting: \.body) // fallback: text
    }
}
Text(note.title).draggable(note)
Color.clear.dropDestination(for: Note.self) { notes, _ in
    inbox += notes; return true
}
ShareLink(item: note, preview: SharePreview(note.title))
```

Security-scoped URLs

```
.fileImporter(isPresented: $open,
              allowedContentTypes: [.pdf]) { r in
    guard case .success(let url) = r,
          url.startAccessingSecurityScopedResource() else { return }
    defer { url.stopAccessingSecurityScopedResource() } // balance!
    let data = try? Data(contentsOf: url)
    saved = try? url.bookmarkData() // persist THIS, not the path
}
```

Next launch: `URL(resolvingBookmarkData:..., bookmarkDataIsStale:)`; `stale` ⇒ re-create the bookmark while you have access. Unbalanced start calls leak kernel resources.

Traps

- ! **Silent drop rejection**: `dropDestination(for: URL.self)` never fires for a photo from Photos — it offers image data, not a file URL. Accept what is actually offered.
- ! **200 MB video as DataRepresentation** — materialised in RAM, memory spike / jetsam. Use `FileRepresentation`.
- ! **ReceivedTransferredFile.file used after the closure** — gone; copy it inside the closure.
- ! **Everyone receives plain text**: the custom UTType is not declared in `Info.plist`, or the text representation is listed first.
- ! **Saving the imported path** instead of a bookmark: works in the callback, fails next launch.

Which representation?

Codable	your own model between your apps / instances
Data	small binary in a standard format (PNG, PDF)
File	anything big, or already on disk (video, documents)
Proxy	"as text / as URL" fallback, no encoding code

Remember

“Describe once, richest first, big things as files, borrowed files by bookmark.”